Effects of Baits and Lures on Camera Trap Detections of East Texas Animals

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Nine-banded armadillo walking toward perfume lure

Introduction

- Baiting and luring are methods of increasing animal activity and detections at camera trapping sites, with potential non-target effects of altering the observed community.
- **Goal:** Identify the effects of baits and lures on detection rates of animal species in the East Texas Pineywoods.
- **Lures and Baits:** Calvin Klein Obsession perfume (known to attract wildcats [Viscarra et al., 2011]) and fish heads were used in this experiment.
- **Expected results:** Fish and perfume sites would have more detections than the control sites. Sites with the perfume lure would attract carnivores (e.g., cats and foxes), sites baited with fish would attract opportunists and scavengers, and both treatments would deter herbivores because of predator avoidance.



White-tailed deer sniffing fish head baited site



Coyote walking towards fish bait

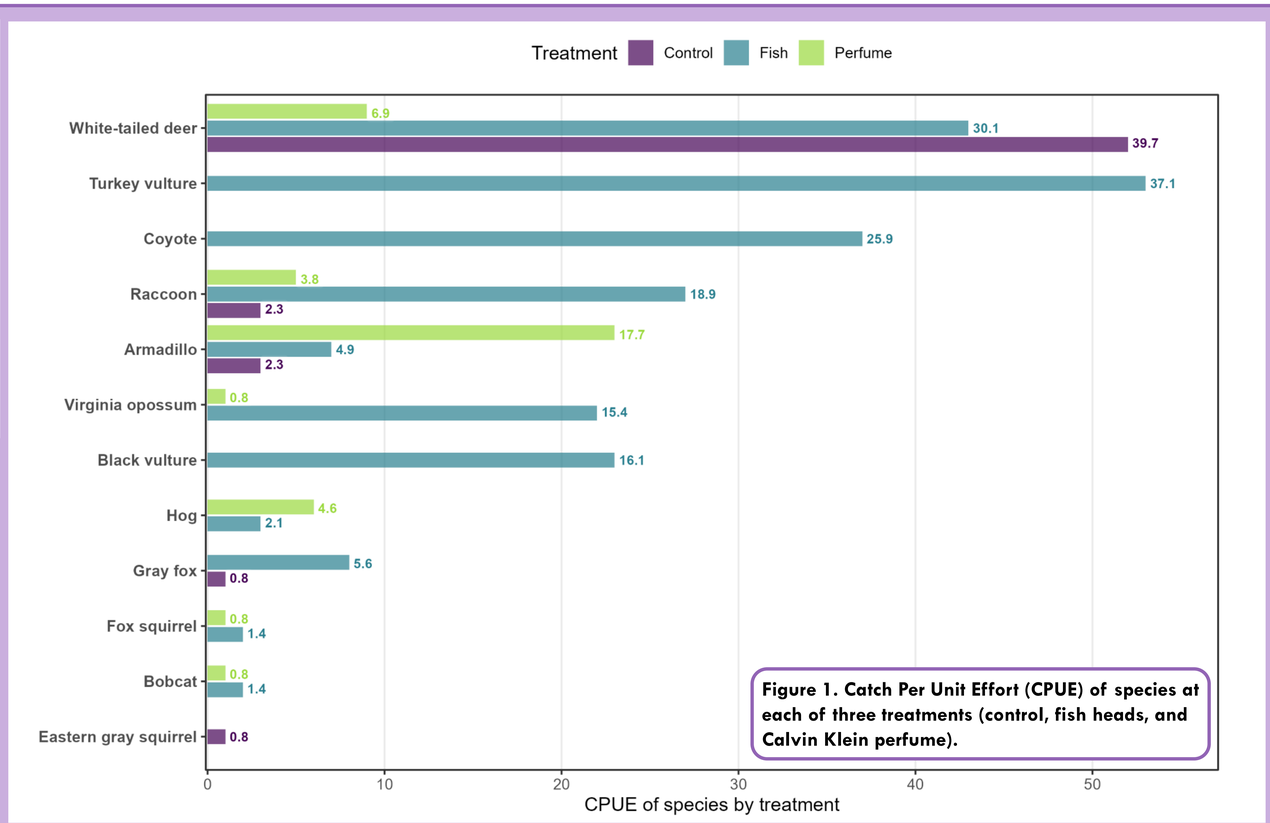


Figure 1. Catch Per Unit Effort (CPUE) of species at each of three treatments (control, fish heads, and Calvin Klein perfume).



Gray fox interacting with fish bait

Methodology

- Twelve camera traps were deployed on the Winston 8 Ranch from June to July 2024
- Cameras were arranged in four triads separated by ≥ 200 meters. Each triad contained three cameras, with one camera assigned to each treatment: fish heads, Calvin Klein Obsession perfume, and no bait (control).
- Catch Per Unit Effort (CPUE) was calculated for each treatment as the number of independent records (<15 minutes) per camera-day x 100 (O'Brien et al. 2003; Iannarilli et al. 2021)

Acknowledgments

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Discussion

- Baits and lures increased detections of individuals and species
- Findings that matched expectations:
  - Coyotes, raccoons, vultures, and Virginia opossums were drawn to fish
- Surprises:
  - White-tailed deer appeared interested in fish
  - Bobcats were not noticeably drawn to perfume—but armadillos were!
- Baiting could be recommended if researchers want to increase certain species detections and improve detection rates
  - Of the treatments tested here, fish was the more effective overall

Results

- Most species detected at fish sites
- Fewest species detected at control sites
- Seven of the twelve total species captured detected only at treated sites

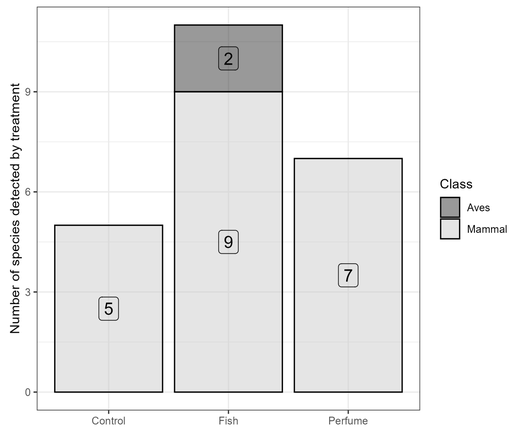


Figure 2. Number of species detected at control, fish, and perfume treatment sites.